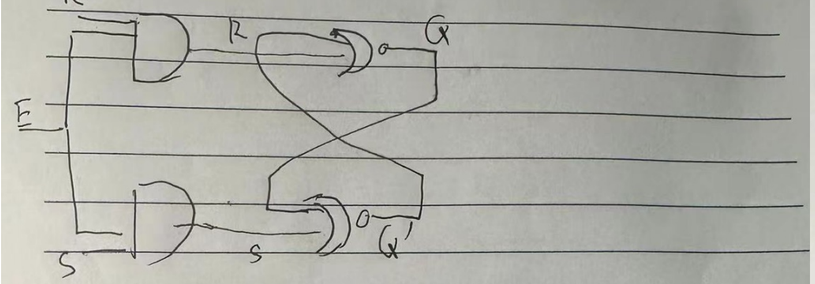
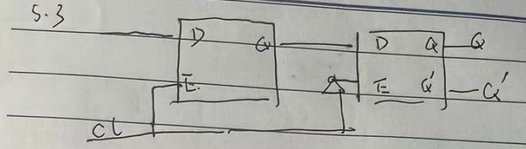
5.1



$\overline{E}$	S	R	Q	Q_{next}	Q'_{next}
0	x	x	0	0	1
0	x	x	1	1	0
1	0	0	0	0	1
1	0	0	1	1	0
1	0	1	x	0	1
1	1	0	x	1	0
1	1	1	x	0	0

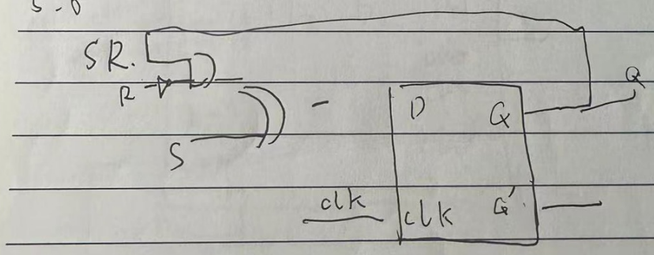
5.3

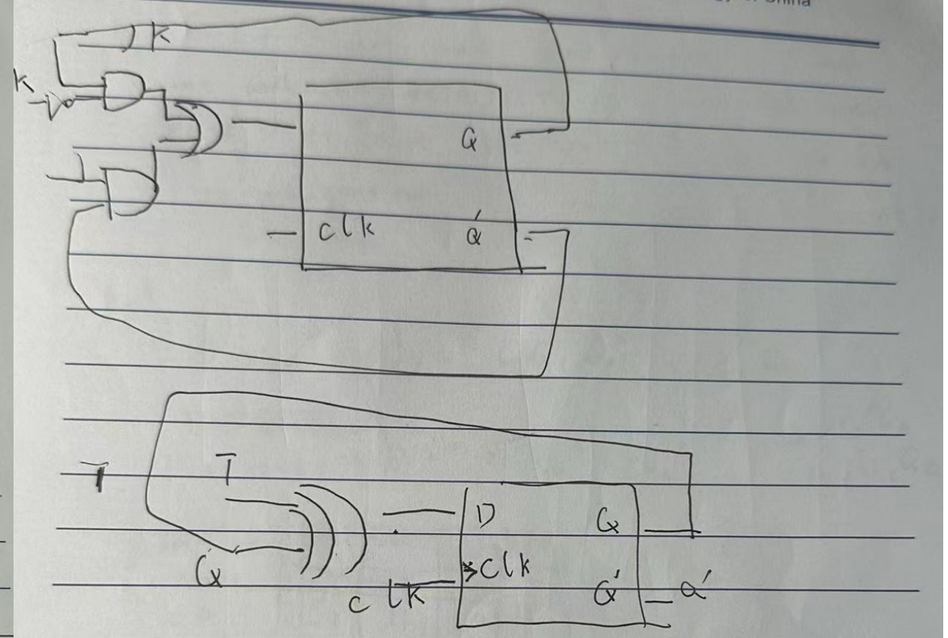


5.4

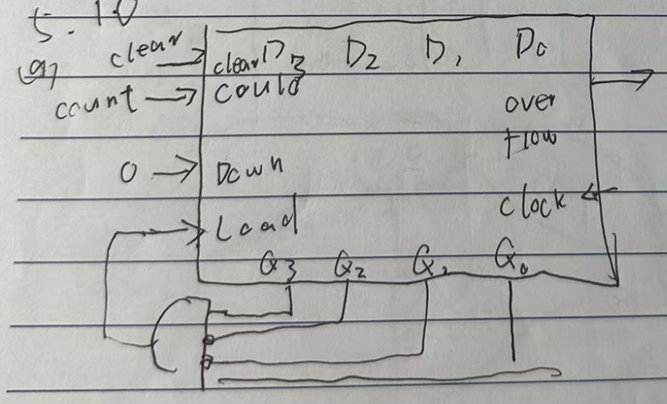
clk	D	Q	Q_{next}	Q_{next}
0	x	0	0	1
0	x	1	1	0
1	x	0	0	1
1	x	1	1	0
↓	0	x	0	1
↓	1	x	1	0

5.8





5.10

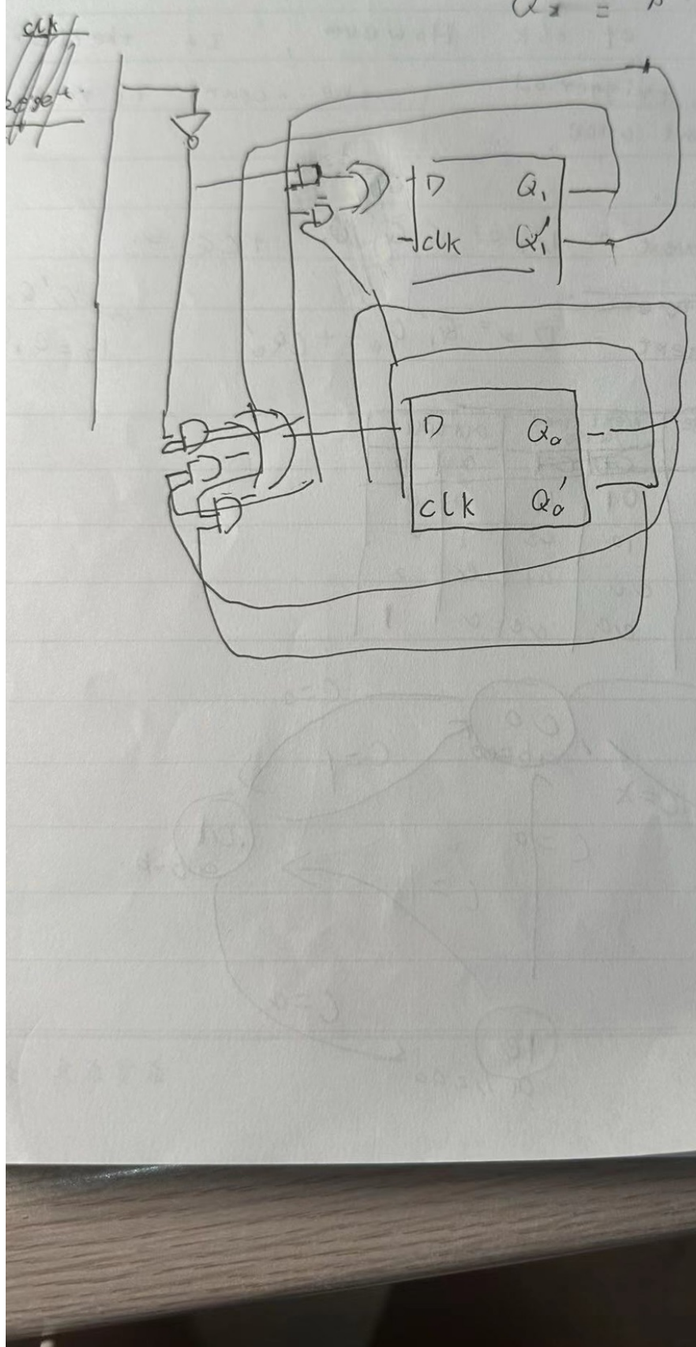


6.3

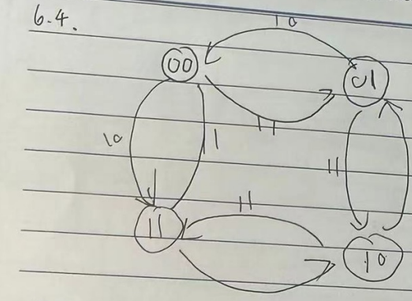
$Q_1^* = A \cdot Q_1'Q_0 + AQ_1'Q_0' + A'Q_1''Q_1'$

$= AQ_1' + Q_1'Q_0'$

$Q_0^* = A'Q_1' + Q_1'Q_0 + Q_1Q_0'$



6.4.

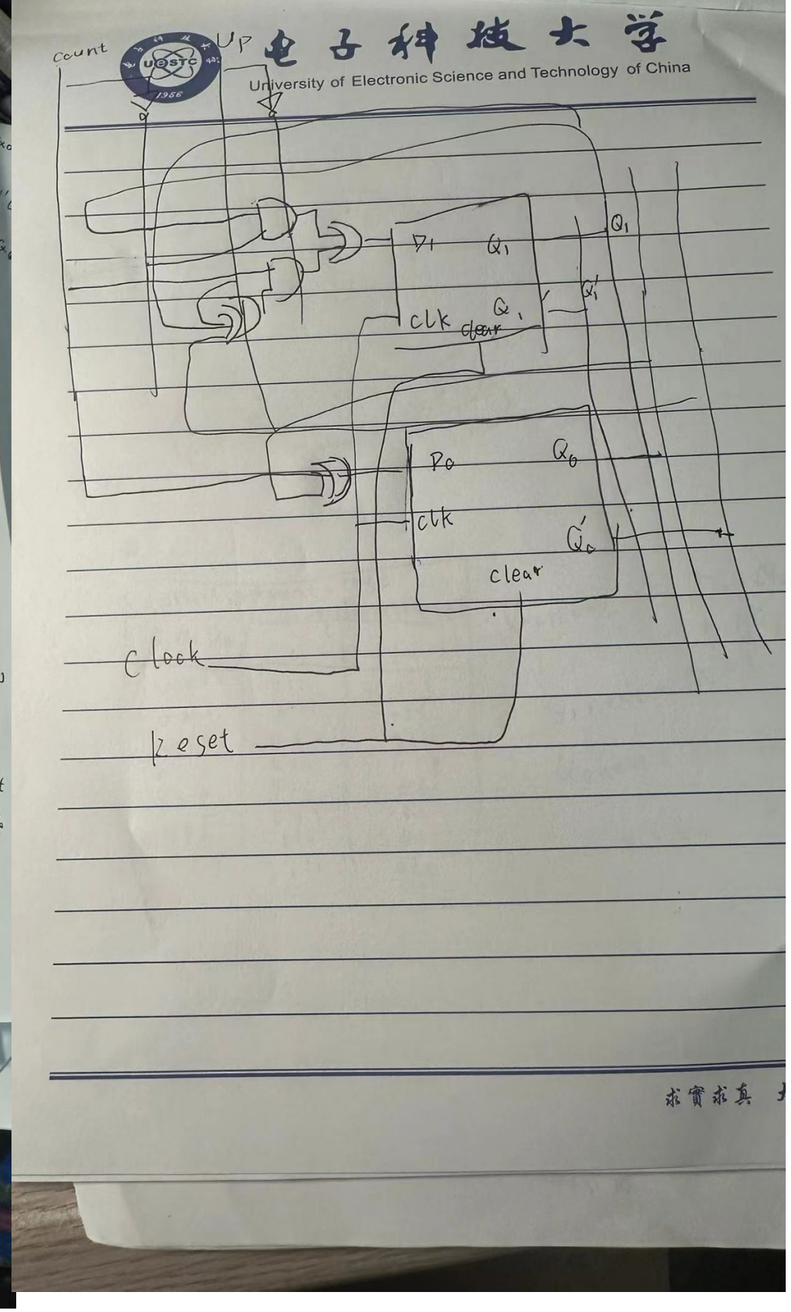


Current state $Q_1\ Q_0$	Next state $Q_{1next}\ Q_{0next}$ count, UP= 00	01	10	11
00	00	01	11	01
01	01	01	00	10
10	10	10	01	11
11	11	11	10	00

$Q_{1next} = D_1 = count'Q_1 + count(Up \oplus Q_1 \oplus Q_0)$

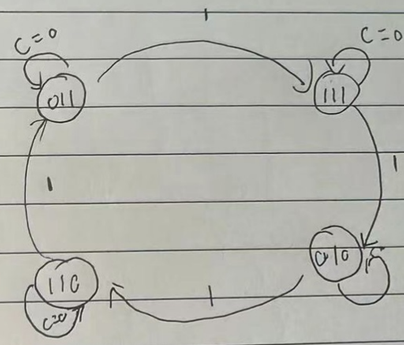
$Q_{0next} = D_0 = count'Q_0 + count(Up \oplus Q_0)' = count \oplus Q_0$

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6.7



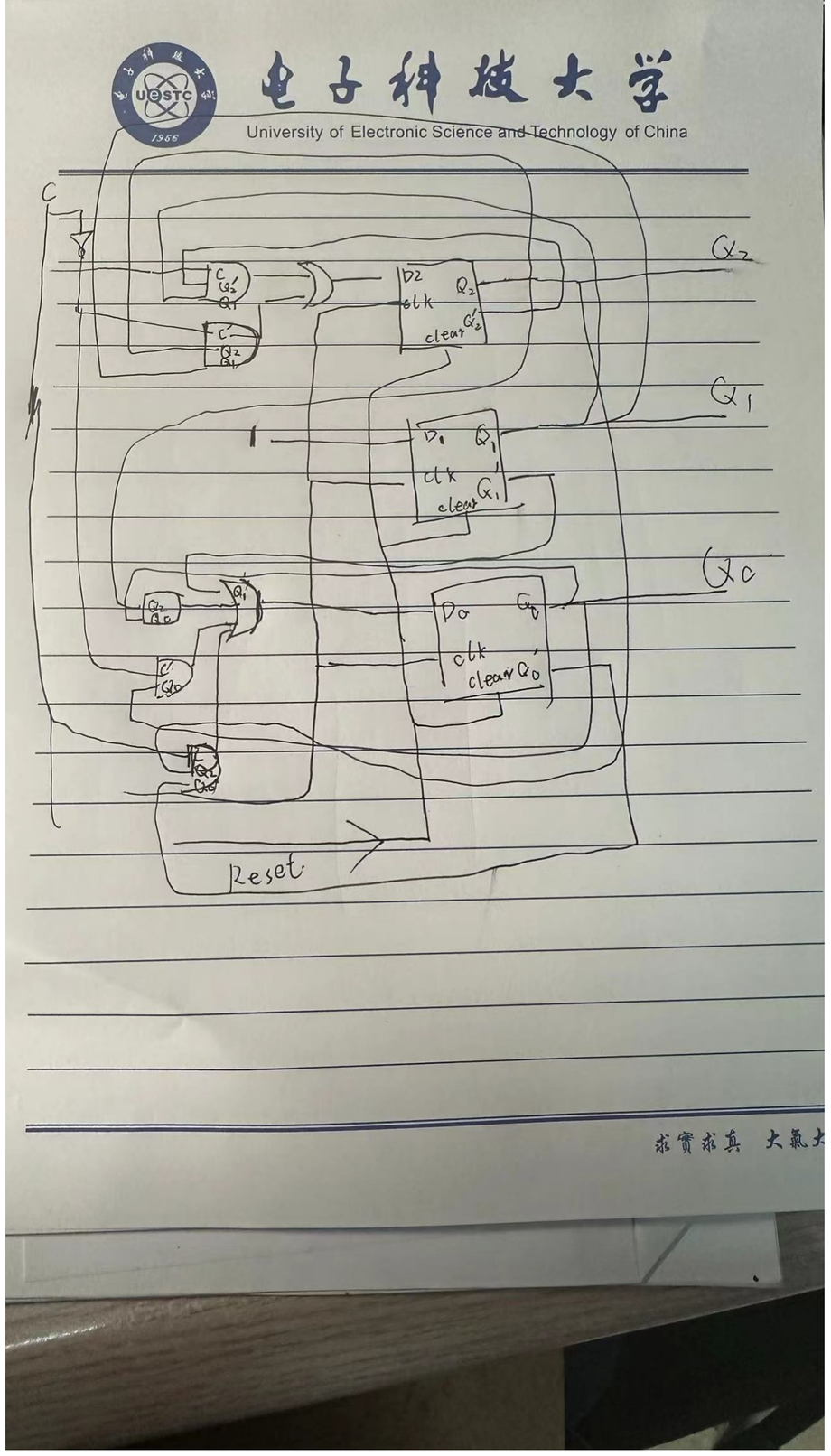
Current stage $Q_2Q_1Q_0$	Next stage $Q_{2next}Q_{1next}Q_{0next}$ C= 0	1
000	011	011
001	011	011
010	010	110
011	010	111
100	011	011
101	110	011
110	111	011
111	111	010

$Q_{2next} = D_2 = Q_2'Q_1 + C'Q_2Q_1$

$Q_{1next} = D_1 = 1$

$Q_{0next} = D_0 = Q_1' + Q_2'C'Q_0 + CQ_2Q_0'$

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